

5 Mid-Level Data Science Projects

1■■■ Customer Churn Prediction

Predict whether a customer will stop using a company's service (like telecom, bank, or SaaS).

Key Skills: Logistic Regression, Random Forest, Feature Engineering

- Import a telecom/banking churn dataset (e.g., Kaggle Telco Churn).
- Clean data — handle missing values and encode categorical variables.
- Perform EDA (correlation, churn vs. tenure, etc.).
- Train-test split and apply classification models.
- Evaluate using ROC-AUC, F1-score, and feature importance.

2■■■ House Price Prediction

Predict house prices based on various features like location, size, number of rooms, etc.

Key Skills: Regression, Feature Scaling, Regularization

- Use the "California Housing" or "Ames Housing" dataset.
- Handle outliers and missing values.
- Encode categorical variables (location, type, etc.).
- Train Linear Regression, Random Forest, and XGBoost.
- Evaluate using RMSE and R^2 ; compare models.

3■■■ Sentiment Analysis on Social Media Tweets

Classify tweets as Positive, Negative, or Neutral.

Key Skills: NLP, Text Preprocessing, TF-IDF, Logistic Regression

- Collect tweets using Twitter API or use a Kaggle dataset.
- Clean text (remove URLs, stopwords, special characters).
- Convert text to numerical form using TF-IDF or Word2Vec.
- Train classification model (e.g., Logistic Regression, Naive Bayes).
- Visualize word clouds and sentiment distribution.

4■■■ Credit Card Fraud Detection

Detect fraudulent transactions using anomaly detection and classification.

Key Skills: Imbalanced Data Handling, Classification, SMOTE

- Use the "Credit Card Fraud Detection" dataset (Kaggle).
- Analyze data imbalance (fraud vs. non-fraud).
- Apply resampling techniques (SMOTE/undersampling).
- Train models — Logistic Regression, Random Forest, XGBoost.
- Evaluate using Precision, Recall, and Confusion Matrix.

5■■■ Movie Recommendation System

Build a system that recommends movies based on user preferences.

Key Skills: Collaborative Filtering, Cosine Similarity, Matrix Factorization

- Use the "MovieLens" dataset.

- Create a user-item rating matrix.
- Implement content-based filtering (using genres/tags).
- Implement collaborative filtering (user-user or item-item).
- Recommend top N movies for a given user and evaluate accuracy.